

# CLamb

## Grammar Specifications & Parser

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### 1. Grammar Specifications

#### 1.1. Start Symbol:

|                 |           |
|-----------------|-----------|
| Starting Symbol | <Program> |
|-----------------|-----------|

#### 1.2. Terminals:

| Terminals              |                                                                                        |
|------------------------|----------------------------------------------------------------------------------------|
| Data Types             | Int, float, char, bool, string, void, auto                                             |
| Keyword if             | if                                                                                     |
| Keyword else           | else                                                                                   |
| Keyword for            | for                                                                                    |
| Keyword while          | while                                                                                  |
| Keyword return         | return                                                                                 |
| Keyword lamb           | lamb                                                                                   |
| Keyword null           | null                                                                                   |
| Keyword print          | print                                                                                  |
| Keyword malloc         | malloc                                                                                 |
| Keyword free           | free                                                                                   |
| Identifiers & Literals | Identifier, integer_literal, float_literal, char_literal, bool_literal, string_literal |
| Operators              | +, -, *, /, %, AND, OR, NOT, <, >, <=, >=, ==, !=, =, &                                |
| Delimiters             | (, ), {, }, [, ], ;, ,                                                                 |

### 1.3. Non-Terminals:

| Non-Terminal         |                   |                  |                      |
|----------------------|-------------------|------------------|----------------------|
| <Program>            | <DeclarationList> | <Declaration>    | <VarDecl>            |
| <FuncDecl>           | <Type>            | <PtrOpt>         | <VarInit>            |
| <ParamList>          | <ParamDecl>       | <ParamTail>      | <Block>              |
| <StmtList>           | <Statement>       | <ExprStmt>       | <IfStmt>             |
| <ElseOpt>            | <WhileStmt>       | <ForStmt>        | <ReturnStmt>         |
| <PrintStmt>          | <FreeStmt>        | <ExprOpt>        | <Expression>         |
| <AssignExpr>         | <AssignTail>      | <LogicalOrExpr>  | <LogicalOrTail>      |
| <LogicalAndExpr>     | <LogicalAndTail>  | <EqualityExpr>   | <EqualityTail>       |
| <EqOp>               | <RelationalExpr>  | <RelationalTail> | <RelOp>              |
| <AdditiveExpr>       | <AdditiveTail>    | <AddOp>          | <MultiplicativeExpr> |
| <MultiplicativeTail> | <MulOp>           | <UnaryExpr>      | <UnaryOp>            |
| <PostfixExpr>        | <PostfixTail>     | <PrimaryExpr>    | <LambdaExpr>         |
| <ExprList>           | <ExprTail>        |                  |                      |

### 1.4. Production Rules:

| Production Rules                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program Structure &amp; Declarations</b> | <p> &lt;Program&gt; ::= &lt;DeclarationList&gt;<br/> &lt;DeclarationList&gt; ::= &lt;Declaration&gt;   &lt;Declaration&gt; &lt;DeclarationList&gt;<br/> &lt;Declaration&gt; ::= &lt;VarDecl&gt;   &lt;FuncDecl&gt; </p> <p> &lt;VarDecl&gt; ::= &lt;Type&gt; &lt;PtrOpt&gt; identifier &lt;VarInit&gt; semicolon<br/> &lt;FuncDecl&gt; ::= &lt;Type&gt; &lt;PtrOpt&gt; identifier l_paren &lt;ParamList&gt; r_paren &lt;Block&gt; </p> <p> &lt;Type&gt; ::= data_type<br/> &lt;PtrOpt&gt; ::= asterisk   ε </p> <p> &lt;VarInit&gt; ::= assignment_op &lt;Expression&gt;   l_bracket integer_literal r_bracket   ε<br/> &lt;ParamList&gt; ::= &lt;ParamDecl&gt; &lt;ParamTail&gt;   ε<br/> &lt;ParamTail&gt; ::= comma &lt;ParamDecl&gt; &lt;ParamTail&gt;   ε </p> |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | $\langle \text{ParamDecl} \rangle ::= \langle \text{Type} \rangle \langle \text{PtrOpt} \rangle \text{identifier}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Statement &amp; Blocks</b>      | $\langle \text{Block} \rangle ::= \text{l\_brace} \langle \text{StmtList} \rangle \text{r\_brace}$<br>$\langle \text{StmtList} \rangle ::= \langle \text{Statement} \rangle \langle \text{StmtList} \rangle \mid \epsilon$<br><br>$\langle \text{Statement} \rangle ::= \langle \text{VarDecl} \rangle$<br>$\quad \mid \langle \text{Block} \rangle$<br>$\quad \mid \langle \text{ExprStmt} \rangle$<br>$\quad \mid \langle \text{IfStmt} \rangle$<br>$\quad \mid \langle \text{WhileStmt} \rangle$<br>$\quad \mid \langle \text{ForStmt} \rangle$<br>$\quad \mid \langle \text{ReturnStmt} \rangle$<br>$\quad \mid \langle \text{PrintStmt} \rangle$<br>$\quad \mid \langle \text{FreeStmt} \rangle$<br><br>$\langle \text{ExprStmt} \rangle ::= \langle \text{Expression} \rangle \text{semicolon} \mid \text{semicolon}$<br>$\langle \text{IfStmt} \rangle ::= \text{kw\_if} \text{l\_paren} \langle \text{Expression} \rangle \text{r\_paren} \langle \text{Block} \rangle \langle \text{ElseOpt} \rangle$<br>$\langle \text{ElseOpt} \rangle ::= \text{kw\_else} \langle \text{Block} \rangle \mid \text{kw\_else} \langle \text{IfStmt} \rangle \mid \epsilon$<br>$\langle \text{WhileStmt} \rangle ::= \text{kw\_while} \text{l\_paren} \langle \text{Expression} \rangle \text{r\_paren} \langle \text{Block} \rangle$<br>$\langle \text{ForStmt} \rangle ::= \text{kw\_for} \text{l\_paren} \langle \text{ExprOpt} \rangle \text{semicolon} \langle \text{ExprOpt} \rangle$<br>$\text{semicolon} \langle \text{ExprOpt} \rangle \text{r\_paren} \langle \text{Block} \rangle$<br>$\langle \text{ReturnStmt} \rangle ::= \text{kw\_return} \langle \text{ExprOpt} \rangle \text{semicolon}$<br>$\langle \text{PrintStmt} \rangle ::= \text{kw\_print} \text{l\_paren} \langle \text{ExprList} \rangle \text{r\_paren} \text{semicolon}$<br>$\langle \text{FreeStmt} \rangle ::= \text{kw\_free} \text{l\_paren} \langle \text{Expression} \rangle \text{r\_paren} \text{semicolon}$                                                                                                                                                                          |
| <b>Expressions &amp; Operators</b> | $\langle \text{ExprOpt} \rangle ::= \langle \text{Expression} \rangle \mid \epsilon$<br>$\langle \text{Expression} \rangle ::= \langle \text{AssignExpr} \rangle$<br><br>$\langle \text{AssignExpr} \rangle ::= \langle \text{LogicalOrExpr} \rangle \langle \text{AssignTail} \rangle$<br>$\langle \text{AssignTail} \rangle ::= \text{assignment\_op} \langle \text{AssignExpr} \rangle \mid \epsilon$<br><br>$\langle \text{LogicalOrExpr} \rangle ::= \langle \text{LogicalAndExpr} \rangle \langle \text{LogicalOrTail} \rangle$<br>$\langle \text{LogicalOrTail} \rangle ::= \text{logical\_or} \langle \text{LogicalAndExpr} \rangle \langle \text{LogicalOrTail} \rangle \mid \epsilon$<br><br>$\langle \text{LogicalAndExpr} \rangle ::= \langle \text{EqualityExpr} \rangle \langle \text{LogicalAndTail} \rangle$<br>$\langle \text{LogicalAndTail} \rangle ::= \text{logical\_and} \langle \text{EqualityExpr} \rangle \langle \text{LogicalAndTail} \rangle \mid \epsilon$<br><br>$\langle \text{EqualityExpr} \rangle ::= \langle \text{RelationalExpr} \rangle \langle \text{EqualityTail} \rangle$<br>$\langle \text{EqualityTail} \rangle ::= \langle \text{EqOp} \rangle \langle \text{RelationalExpr} \rangle \langle \text{EqualityTail} \rangle \mid \epsilon$<br><br>$\langle \text{EqOp} \rangle ::= \text{relational\_eq} \mid \text{relational\_noteq}$<br><br>$\langle \text{RelationalExpr} \rangle ::= \langle \text{AdditiveExpr} \rangle \langle \text{RelationalTail} \rangle$<br>$\langle \text{RelationalTail} \rangle ::= \langle \text{RelOp} \rangle \langle \text{AdditiveExpr} \rangle \langle \text{RelationalTail} \rangle \mid \epsilon$<br><br>$\langle \text{RelOp} \rangle ::= \text{relational\_greater} \mid \text{relational\_less} \mid \text{relational\_lesseq}$<br>$\quad \mid \text{relational\_greatereq}$<br><br>$\langle \text{AdditiveExpr} \rangle ::= \langle \text{MultiplicativeExpr} \rangle \langle \text{AdditiveTail} \rangle$<br>$\langle \text{AdditiveTail} \rangle ::= \langle \text{AddOp} \rangle \langle \text{MultiplicativeExpr} \rangle \langle \text{AdditiveTail} \rangle \mid \epsilon$ |

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <pre> &lt;AddOp&gt; ::= add_op  &lt;MultiplicativeExpr&gt; ::= &lt;UnaryExpr&gt; &lt;MultiplicativeTail&gt; &lt;MultiplicativeTail&gt; ::= &lt;MulOp&gt; &lt;UnaryExpr&gt; &lt;MultiplicativeTail&gt;   ε  &lt;MulOp&gt; ::= mul_op   asterisk  &lt;UnaryExpr&gt; ::= &lt;UnaryOp&gt; &lt;UnaryExpr&gt;   &lt;PostfixExpr&gt; &lt;UnaryOp&gt; ::= logical_not   address_op   asterisk   add_op  &lt;PostfixExpr&gt; ::= &lt;PrimaryExpr&gt; &lt;PostfixTail&gt; &lt;PostfixTail&gt; ::= l_bracket &lt;Expression&gt; r_bracket &lt;PostfixTail&gt;                   l_paren &lt;ExprList&gt; r_paren &lt;PostfixTail&gt;                   ε </pre> |
| <b>Primary Expressions &amp; Lambdas</b> | <pre> &lt;PrimaryExpr&gt; ::= identifier                   integer_literal                   float_literal                   char_literal                   string_literal                   boolean_literal                   kw_null                   l_paren &lt;Expression&gt; r_paren                   &lt;LambdaExpr&gt;                   kw_malloc l_paren &lt;Expression&gt; r_paren  &lt;LambdaExpr&gt; ::= kw_lamb l_paren &lt;ParamList&gt; r_paren &lt;Block&gt;  &lt;ExprList&gt; ::= &lt;Expression&gt; &lt;ExprTail&gt;   ε &lt;ExprTail&gt; ::= comma &lt;Expression&gt; &lt;ExprTail&gt;   ε </pre>                              |

## 2. LL(k) Grammar

Here is our modified LL(1) version of the grammar. The entire grammar has been made non=left-recursive. The statements that have changed/added are highlighted in red, and the statements that are removed are also highlighted in red and explicitly mentioned in the same red colour.

| Production Rules                            |                                                                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program Structure &amp; Declarations</b> | <pre> &lt;Program&gt; ::= &lt;DeclarationList&gt;  &lt;DeclarationList&gt; ::= &lt;Declaration&gt; &lt;DeclarationListTail&gt;   ε &lt;DeclarationListTail&gt; ::= &lt;Declaration&gt; &lt;DeclarationListTail&gt;   ε </pre> |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p> <code>&lt;Declaration&gt; ::= data_type &lt;PtrOpt&gt; identifier &lt;DeclTail&gt;</code><br/> <code>&lt;DeclTail&gt; ::= &lt;VarInit&gt; semicolon   l_paren &lt;ParamList&gt; r_paren &lt;Block&gt;</code> </p> <p> <code>&lt;PtrOpt&gt; ::= asterisk   ε</code><br/> <code>&lt;VarInit&gt; ::= assignment_op &lt;Expression&gt;   l_bracket integer_literal r_bracket   ε</code> </p> <p> <code>&lt;ParamList&gt; ::= &lt;ParamDecl&gt; &lt;ParamTail&gt;   ε</code><br/> <code>&lt;ParamTail&gt; ::= comma &lt;ParamDecl&gt; &lt;ParamTail&gt;   ε</code><br/> <code>&lt;ParamDecl&gt; ::= data_type &lt;PtrOpt&gt; identifier</code> </p> <p> ( <br/> Removed <code>&lt;Type&gt; ::= data_type</code><br/> Removed FIRST FIRST conflict in DeclarationList and Declaration by adding tails<br/> Removed VarDecl and FuncDecl<br/> ) </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Statement &amp; Blocks</b> | <p> <code>&lt;Block&gt; ::= l_brace &lt;StmtList&gt; r_brace</code><br/> <code>&lt;StmtList&gt; ::= &lt;Statement&gt; &lt;StmtList&gt;   ε</code> </p> <p> <code>&lt;Statement&gt; ::= &lt;Declaration&gt;</code><br/>   <code>&lt;Block&gt;</code><br/>   <code>&lt;ExprStmt&gt;</code><br/>   <code>&lt;IfStmt&gt;</code><br/>   <code>&lt;WhileStmt&gt;</code><br/>   <code>&lt;ForStmt&gt;</code><br/>   <code>&lt;ReturnStmt&gt;</code><br/>   <code>&lt;PrintStmt&gt;</code><br/>   <code>&lt;FreeStmt&gt;</code> </p> <p> <code>&lt;ExprStmt&gt; ::= &lt;Expression&gt; semicolon   semicolon</code> </p> <p> <code>&lt;IfStmt&gt; ::= kw_if l_paren &lt;Expression&gt; r_paren &lt;Block&gt; &lt;ElseOpt&gt;</code><br/> <code>&lt;ElseOpt&gt; ::= kw_else &lt;ElseBody&gt;   ε</code><br/> <code>&lt;ElseBody&gt; ::= &lt;Block&gt;   &lt;IfStmt&gt;</code> </p> <p> <code>&lt;WhileStmt&gt; ::= kw_while l_paren &lt;Expression&gt; r_paren &lt;Block&gt;</code><br/> <code>&lt;ForStmt&gt; ::= kw_for l_paren &lt;ExprOpt&gt; semicolon &lt;ExprOpt&gt; semicolon &lt;ExprOpt&gt; r_paren &lt;Block&gt;</code><br/> <code>&lt;ReturnStmt&gt; ::= kw_return &lt;ExprOpt&gt; semicolon</code><br/> <code>&lt;PrintStmt&gt; ::= kw_print l_paren &lt;ExprList&gt; r_paren semicolon</code><br/> <code>&lt;FreeStmt&gt; ::= kw_free l_paren &lt;Expression&gt; r_paren semicolon</code> </p> <p> ( <br/> Replaced VarDecl in Statement with Declaration as it now directly handles both variable and function declarations.<br/> This does however, allow nested functions in the language. This can be handled in the semantics stage of the language.<br/> ) </p> |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | <p>Removed FIRST FIRST conflict in the “Else”<br/>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Expressions &amp; Operators</b> | <p> <math>\langle \text{ExprOpt} \rangle ::= \langle \text{Expression} \rangle \mid \epsilon</math><br/> <math>\langle \text{Expression} \rangle ::= \langle \text{AssignExpr} \rangle</math> </p> <p> <math>\langle \text{AssignExpr} \rangle ::= \langle \text{LogicalOrExpr} \rangle \langle \text{AssignTail} \rangle</math><br/> <math>\langle \text{AssignTail} \rangle ::= \text{assignment\_op} \langle \text{AssignExpr} \rangle \mid \epsilon</math> </p> <p> <math>\langle \text{LogicalOrExpr} \rangle ::= \langle \text{LogicalAndExpr} \rangle \langle \text{LogicalOrTail} \rangle</math><br/> <math>\langle \text{LogicalOrTail} \rangle ::= \text{logical\_or} \langle \text{LogicalAndExpr} \rangle \langle \text{LogicalOrTail} \rangle \mid \epsilon</math> </p> <p> <math>\langle \text{LogicalAndExpr} \rangle ::= \langle \text{EqualityExpr} \rangle \langle \text{LogicalAndTail} \rangle</math><br/> <math>\langle \text{LogicalAndTail} \rangle ::= \text{logical\_and} \langle \text{EqualityExpr} \rangle \langle \text{LogicalAndTail} \rangle \mid \epsilon</math> </p> <p> <math>\langle \text{EqualityExpr} \rangle ::= \langle \text{RelationalExpr} \rangle \langle \text{EqualityTail} \rangle</math><br/> <math>\langle \text{EqualityTail} \rangle ::= \text{relational\_eq} \langle \text{RelationalExpr} \rangle \langle \text{EqualityTail} \rangle</math><br/> <math>\quad \mid \text{relational\_noteq} \langle \text{RelationalExpr} \rangle \langle \text{EqualityTail} \rangle</math><br/> <math>\quad \mid \epsilon</math> </p> <p> <math>\langle \text{RelationalExpr} \rangle ::= \langle \text{AdditiveExpr} \rangle \langle \text{RelationalTail} \rangle</math><br/> <math>\langle \text{RelationalTail} \rangle ::= \text{relational\_less} \langle \text{AdditiveExpr} \rangle \langle \text{RelationalTail} \rangle</math><br/> <math>\quad \mid \text{relational\_greater} \langle \text{AdditiveExpr} \rangle \langle \text{RelationalTail} \rangle</math><br/> <math>\quad \mid \text{relational\_lesseq} \langle \text{AdditiveExpr} \rangle \langle \text{RelationalTail} \rangle</math><br/> <math>\quad \mid \text{relational\_greatereq} \langle \text{AdditiveExpr} \rangle \langle \text{RelationalTail} \rangle</math><br/> <math>\quad \mid \epsilon</math> </p> <p> <math>\langle \text{AdditiveExpr} \rangle ::= \langle \text{MultiplicativeExpr} \rangle \langle \text{AdditiveTail} \rangle</math><br/> <math>\langle \text{AdditiveTail} \rangle ::= \text{add\_op} \langle \text{MultiplicativeExpr} \rangle \langle \text{AdditiveTail} \rangle \mid \epsilon</math> </p> <p> <math>\langle \text{MultiplicativeExpr} \rangle ::= \langle \text{UnaryExpr} \rangle \langle \text{MultiplicativeTail} \rangle</math><br/> <math>\langle \text{MultiplicativeTail} \rangle ::= \langle \text{MulOp} \rangle \langle \text{UnaryExpr} \rangle \langle \text{MultiplicativeTail} \rangle \mid \epsilon</math><br/> <math>\langle \text{MulOp} \rangle ::= \text{mul\_op} \mid \text{asterisk}</math> </p> <p> <math>\langle \text{UnaryExpr} \rangle ::= \langle \text{UnaryOp} \rangle \langle \text{UnaryExpr} \rangle \mid \langle \text{PostfixExpr} \rangle</math><br/> <math>\langle \text{UnaryOp} \rangle ::= \text{logical\_not} \mid \text{address\_op} \mid \text{asterisk} \mid \text{add\_op}</math> </p> <p> <math>\langle \text{PostfixExpr} \rangle ::= \langle \text{PrimaryExpr} \rangle \langle \text{PostfixTail} \rangle</math><br/> <math>\langle \text{PostfixTail} \rangle ::= \text{l\_bracket} \langle \text{Expression} \rangle \text{r\_bracket} \langle \text{PostfixTail} \rangle</math><br/> <math>\quad \mid \text{l\_paren} \langle \text{ExprList} \rangle \text{r\_paren} \langle \text{PostfixTail} \rangle</math><br/> <math>\quad \mid \epsilon</math> </p> <p> ( Removed<br/> <math>\langle \text{RelOp} \rangle ::= \text{relational\_op}</math><br/> <math>\langle \text{EqOp} \rangle ::= \text{relational\_op}</math><br/> <math>\langle \text{AddOp} \rangle ::= \text{add\_op}</math> ) </p> |
| <b>Primary</b>                     | $\langle \text{PrimaryExpr} \rangle ::= \text{identifier}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Expressions &amp; Lambdas</b> | integer_literal<br>  float_literal<br>  char_literal<br>  string_literal<br>  boolean_literal<br>  kw_null<br>  l_paren <Expression> r_paren<br>  kw_lamb l_paren <ParamList> r_paren <Block><br>  kw_malloc l_paren <Expression> r_paren<br><br><ExprList> ::= <Expression> <ExprTail>   ε<br><ExprTail> ::= comma <Expression> <ExprTail>   ε<br><br>(<br>Removed<br><LambdaExpr> ::= kw_lamb l_paren <ParamList> r_paren <Block><br>And added it directly into primary expression as it is of fixed format<br>) |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3. First, Follow, & Lookahead

#### 3.1. First & Follow

**FIRST\_EXPR** = {logical\_not, address\_op, asterisk, add\_op, identifier, integer\_literal, float\_literal, char\_literal, string\_literal, boolean\_literal, kw\_null, l\_paren, kw\_lamb, kw\_malloc}

**FIRST\_STMT** = {data\_type, l\_brace, kw\_if, kw\_while, kw\_for, kw\_return, kw\_print, kw\_free, semicolon} ∪ FIRST\_EXPR

| Non-Terminal          | First                       | Follow                  |
|-----------------------|-----------------------------|-------------------------|
| <Program>             | data_type, ε                | \$                      |
| <DeclarationList>     | data_type, ε                | \$                      |
| <DeclarationListTail> | data_type, ε                | \$                      |
| <Declaration>         | data_type                   | FIRST_STMT, r_brace, \$ |
| <PtrOpt>              | asterisk, ε                 | identifier              |
| <VarInit>             | assignment_op, l_bracket, ε | semicolon               |
| <ParamList>           | data_type, ε                | r_paren                 |

|                  |                                              |                                                                              |
|------------------|----------------------------------------------|------------------------------------------------------------------------------|
| <ParamTail>      | comma, $\epsilon$                            | r_paren                                                                      |
| <ParamDecl>      | data_type                                    | comma, r_paren                                                               |
| <DeclTail>       | assignment_op, l_bracket, semicolon, l_paren | FIRST_STMT, r_brace, \$                                                      |
| <ElseBody>       | l_brace, kw_if                               | FIRST_STMT, r_brace                                                          |
| <Block>          | l_brace                                      | FIRST_STMT, r_brace, kw_else, \$                                             |
| <StmtList>       | FIRST_STMT, $\epsilon$                       | r_brace                                                                      |
| <Statement>      | FIRST_STMT                                   | FIRST_STMT, r_brace                                                          |
| <IfStmt>         | kw_if                                        | FIRST_STMT, r_brace                                                          |
| <ElseOpt>        | kw_else, $\epsilon$                          | FIRST_STMT, r_brace                                                          |
| <ExprOpt>        | FIRST_EXPR                                   | semicolon, r_paren                                                           |
| <Expression>     | FIRST_EXPR                                   | semicolon, r_paren, r_bracket, comma                                         |
| <AssignExpr>     | FIRST_EXPR                                   | semicolon, r_paren, r_bracket, comma                                         |
| <AssignTail>     | assignment_op, $\epsilon$                    | semicolon, r_paren, r_bracket, comma                                         |
| <LogicalOrExpr>  | FIRST_EXPR                                   | assignment_op, semicolon, r_paren, r_bracket, comma                          |
| <LogicalOrTail>  | logical_or, $\epsilon$                       | assignment_op, semicolon, r_paren, r_bracket, comma                          |
| <LogicalAndExpr> | FIRST_EXPR                                   | logical_or, assignment_op, semicolon, r_paren, r_bracket, comma              |
| <LogicalAndTail> | logical_and, $\epsilon$                      | logical_or, assignment_op, semicolon, r_paren, r_bracket, comma              |
| <EqualityExpr>   | FIRST_EXPR                                   | logical_and, logical_or, assignment_op, semicolon, r_paren, r_bracket, comma |
| <EqualityTail>   | relational_eq, relational_noteq, $\epsilon$  | logical_and, logical_or, assignment_op, semicolon, r_paren, r_bracket, comma |

|                      |                                                                                                   |                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <RelationalExpr>     | FIRST_EXPR                                                                                        | relational_eq,<br>relational_noteq, logical_and,<br>logical_or, assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma                                                                                                   |
| <RelationalTail>     | relational_less,<br>relational_greater,<br>relational_lesseq,<br>relational_greatereq, $\epsilon$ | relational_eq,<br>relational_noteq, logical_and,<br>logical_or, assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma                                                                                                   |
| <AdditiveExpr>       | FIRST_EXPR                                                                                        | relational_less,<br>relational_greater,<br>relational_lesseq,<br>relational_greatereq,<br>relational_eq,<br>relational_noteq, logical_and,<br>logical_or, assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma         |
| <AdditiveTail>       | add_op, $\epsilon$                                                                                | relational_less,<br>relational_greater,<br>relational_lesseq,<br>relational_greatereq,<br>relational_eq,<br>relational_noteq, logical_and,<br>logical_or, assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma         |
| <MultiplicativeExpr> | FIRST_EXPR                                                                                        | add_op, relational_less,<br>relational_greater,<br>relational_lesseq,<br>relational_greatereq,<br>relational_eq,<br>relational_noteq, logical_and,<br>logical_or, assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma |
| <MultiplicativeTail> | mul_op, asterisk, $\epsilon$                                                                      | add_op, relational_less,<br>relational_greater,<br>relational_lesseq,<br>relational_greatereq,<br>relational_eq,<br>relational_noteq, logical_and,<br>logical_or, assignment_op,<br>semicolon, r_paren,                     |

|               |                                |                                                                                                                                                                                                                                           |
|---------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                                | r_bracket, comma                                                                                                                                                                                                                          |
| <UnaryExpr>   | FIRST_EXPR                     | mul_op, asterisk, add_op, relational_less, relational_greater, relational_lesseq, relational_greatereq, relational_eq, relational_noteq, logical_and, logical_or, assignment_op, semicolon, r_paren, r_bracket, comma                     |
| <PostfixExpr> | FIRST_EXPR                     | mul_op, asterisk, add_op, relational_less, relational_greater, relational_lesseq, relational_greatereq, relational_eq, relational_noteq, logical_and, logical_or, assignment_op, semicolon, r_paren, r_bracket, comma                     |
| <PostfixTail> | l_bracket, l_paren, $\epsilon$ | mul_op, asterisk, add_op, relational_less, relational_greater, relational_lesseq, relational_greatereq, relational_eq, relational_noteq, logical_and, logical_or, assignment_op, semicolon, r_paren, r_bracket, comma                     |
| <PrimaryExpr> | FIRST_EXPR                     | l_bracket, l_paren, mul_op, asterisk, add_op, relational_less, relational_greater, relational_lesseq, relational_greatereq, relational_eq, relational_noteq, logical_and, logical_or, assignment_op, semicolon, r_paren, r_bracket, comma |
| <ExprList>    | FIRST_EXPR, $\epsilon$         | r_paren                                                                                                                                                                                                                                   |
| <MulOp>       | mul_op, asterisk               | FIRST_EXPR                                                                                                                                                                                                                                |

|            |                                           |            |
|------------|-------------------------------------------|------------|
| <UnaryOp>  | logical_not, address_op, asterisk, add_op | FIRST_EXPR |
| <ExprTail> | comma, $\epsilon$                         | r_paren    |

### 3.2. Lookahead

| Type                               | Production                                                       | Lookahead set                       |
|------------------------------------|------------------------------------------------------------------|-------------------------------------|
| <b>Structure &amp; Declaration</b> | <Program> ::= <DeclarationList>                                  | data_type, \$                       |
|                                    | <DeclarationList> ::= <Declaration><br><DeclarationListTail>     | data_type                           |
|                                    | <DeclarationList> ::= $\epsilon$                                 | \$                                  |
|                                    | <DeclarationListTail> ::= <Declaration><br><DeclarationListTail> | data_type                           |
|                                    | <DeclarationListTail> ::= $\epsilon$                             | \$                                  |
|                                    | <Declaration> ::= data_type <PtrOpt> identifier<br><DeclTail>    | data_type                           |
|                                    | <DeclTail> ::= <VarInit> semicolon                               | assignment_op, l_bracket, semicolon |
|                                    | <DeclTail> ::= l_paren <ParamList> r_paren<br><Block>            | l_paren                             |
|                                    | <PtrOpt> ::= asterisk                                            | asterisk                            |
|                                    | <PtrOpt> ::= $\epsilon$                                          | identifier                          |
|                                    | <VarInit> ::= assignment_op <Expression>                         | assignment_op                       |
|                                    | <VarInit> ::= l_bracket integer_literal r_bracket                | l_bracket                           |
|                                    | <VarInit> ::= $\epsilon$                                         | semicolon                           |
| <b>Statements &amp; Blocks</b>     | <StmtList> ::= <Statement> <StmtList>                            | FIRST_STMT                          |
|                                    | <StmtList> ::= $\epsilon$                                        | r_brace                             |
|                                    | <ElseOpt> ::= kw_else <ElseBody>                                 | kw_else                             |
|                                    | <ElseOpt> ::= $\epsilon$                                         | FIRST_STMT, r_brace                 |

|                         |                                                                              |                                                                                          |
|-------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|                         | <ElseBody> ::= <Block>                                                       | l_brace                                                                                  |
|                         | <ElseBody> ::= <IfStmt>                                                      | kw_if                                                                                    |
|                         | <ExprOpt> ::= $\epsilon$                                                     | semicolon, r_paren                                                                       |
|                         | <ExprOpt> ::= <Expression>                                                   | FIRST_EXPR                                                                               |
| <b>Expression Tails</b> | <AssignTail> ::= assignment_op <AssignExpr>                                  | assignment_op                                                                            |
|                         | <AssignTail> ::= $\epsilon$                                                  | semicolon, r_paren,<br>r_bracket, comma                                                  |
|                         | <LogicalOrTail> ::= logical_or<br><LogicalAndExpr> <LogicalOrTail>           | logical_or                                                                               |
|                         | <LogicalOrTail> ::= $\epsilon$                                               | assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma                                |
|                         | <LogicalAndTail> ::= logical_and<br><EqualityExpr> <LogicalAndTail>          | logical_and                                                                              |
|                         | <LogicalAndTail> ::= $\epsilon$                                              | logical_or,<br>assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma                 |
|                         | <EqualityTail> ::= relational_eq<br><RelationalExpr> <EqualityTail>          | relational_eq                                                                            |
|                         | <EqualityTail> ::= relational_noteq<br><RelationalExpr> <EqualityTail>       | relational_noteq                                                                         |
|                         | <EqualityTail> ::= $\epsilon$                                                | logical_and,<br>logical_or,<br>assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma |
|                         | <RelationalTail> ::= relational_less<br><AdditiveExpr> <RelationalTail>      | relational_less                                                                          |
|                         | <RelationalTail> ::= relational_greater<br><AdditiveExpr> <RelationalTail>   | relational_greater                                                                       |
|                         | <RelationalTail> ::= relational_lesseq<br><AdditiveExpr> <RelationalTail>    | relational_lesseq                                                                        |
|                         | <RelationalTail> ::= relational_greatereq<br><AdditiveExpr> <RelationalTail> | relational_greatereq                                                                     |

|  |                                                                      |                                                                                                                                                                                                                                      |
|--|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <RelationalTail> ::= $\epsilon$                                      | relational_eq,<br>relational_noteq,<br>logical_and,<br>logical_or,<br>assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma                                                                                                      |
|  | <AdditiveTail> ::= add_op <MultiplicativeExpr><br><AdditiveTail>     | add_op                                                                                                                                                                                                                               |
|  | <AdditiveTail> ::= $\epsilon$                                        | relational_less,<br>relational_greater,<br>relational_lesseq,<br>relational_greatereq,<br>relational_eq,<br>relational_noteq,<br>logical_and,<br>logical_or,<br>assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma            |
|  | <MultiplicativeTail> ::= <MulOp> <UnaryExpr><br><MultiplicativeTail> | mul_op, asterisk                                                                                                                                                                                                                     |
|  | <MultiplicativeTail> ::= $\epsilon$                                  | add_op,<br>relational_less,<br>relational_greater,<br>relational_lesseq,<br>relational_greatereq,<br>relational_eq,<br>relational_noteq,<br>logical_and,<br>logical_or,<br>assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma |
|  | <PostfixTail> ::= l_bracket <Expression><br>r_bracket <PostfixTail>  | l_bracket                                                                                                                                                                                                                            |
|  | <PostfixTail> ::= l_paren <ExprList> r_paren<br><PostfixTail>        | l_paren                                                                                                                                                                                                                              |
|  | <PostfixTail> ::= $\epsilon$                                         | mul_op, asterisk,<br>add_op,<br>relational_less,<br>relational_greater,<br>relational_lesseq,<br>relational_greatereq,                                                                                                               |

|  |  |                                                                                                                                 |
|--|--|---------------------------------------------------------------------------------------------------------------------------------|
|  |  | relational_eq,<br>relational_noteq,<br>logical_and,<br>logical_or,<br>assignment_op,<br>semicolon, r_paren,<br>r_bracket, comma |
|--|--|---------------------------------------------------------------------------------------------------------------------------------|